

Question	Marking details	Marks Available																	
		AO1	AO2	AO3	Total	Maths	Prac												
6	Indicative content	4	5	0	9	0	0												
	<u>Structure of water:</u>																		
	Water has the chemical formula of H ₂ O.																		
	Water consists of one oxygen atom joined to two hydrogen atoms by covalent bonds.																		
	The molecule is a dipole.																		
	The oxygen atom has a slight negative charge / δ-, the hydrogen atoms have a slightly positive charge δ+.																		
	<table><tr><th><u>Properties of water</u></th><th><u>Essential to supporting life</u></th></tr><tr><td>Ice is less dense than water</td><td>Ice floats insulating the water below allowing organisms to survive.</td></tr><tr><td>Polar solvent</td><td>All chemical reactions occur in aqueous solutions. Polar molecules and ions can dissolve and therefore be transported</td></tr><tr><td>Metabolite</td><td>Involved in chemical reactions</td></tr><tr><td>High specific heat capacity</td><td>Requires a lot of energy to cause water to heat up / cool down therefore creating thermostable environments</td></tr><tr><td>Cohesive / forms hydrogen bonds</td><td>Water molecules form hydrogen bonds allowing the movement of water</td></tr></table>							<u>Properties of water</u>	<u>Essential to supporting life</u>	Ice is less dense than water	Ice floats insulating the water below allowing organisms to survive.	Polar solvent	All chemical reactions occur in aqueous solutions. Polar molecules and ions can dissolve and therefore be transported	Metabolite	Involved in chemical reactions	High specific heat capacity	Requires a lot of energy to cause water to heat up / cool down therefore creating thermostable environments	Cohesive / forms hydrogen bonds	Water molecules form hydrogen bonds allowing the movement of water
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<u>Chemical elements:</u>																			
Phosphorous is required to synthesise phospholipids / nucleotides / ATP / DNA / RNA																			
Sulfur is required to synthesise some amino acids / methionine / cysteine / proteins.																			

			7-9 marks Indicative content of this level is detailed description of all three areas of content <i>The candidate constructs an articulate, integrated account, correctly linking relevant points, such as those in the indicative content, which shows sequential reasoning. The answer fully addresses the question with no irrelevant inclusions or significant omissions. The candidate uses scientific conventions and vocabulary appropriately and accurately.</i> 4-6 marks Indicative content of this level is detailed description of two areas or less description of all three areas <i>The candidate constructs an account correctly linking some relevant points, such as those in the indicative content, showing some reasoning. The answer addresses the question with some omissions. The candidate usually uses scientific conventions and vocabulary appropriately and accurately.</i> 1-3 marks Indicative content of this level is any relevant indicative comment <i>The candidate makes some relevant points, such as those in the indicative content, showing limited reasoning. The answer addresses the question with significant omissions. The candidate has limited use of scientific conventions and vocabulary.</i> 0 marks <i>The candidate does not make any attempt or give a relevant answer worthy of credit.</i>						
			Question 6 total	4	5	0	9	0	0